

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12407724

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Gorter; Charlene Moss et al.

Analyzing messages for malicious content using a cloud computing system

Abstract

A method for analyzing messages for malicious content is provided. The method includes intercepting a message comprising a quick response (QR) code or a uniform resource locator (URL). The QR code corresponds to the URL. A status of the message is determined by determining a source of the message. The source of the message is compared to known malicious sources. In response to the source of the message not matching any of the known malicious sources, the source of the message is compared to known safe sources. In response to the source of the message not matching any of the known safe sources, the URL is compared to known malicious URLs. In response to the URL matching a respective one of the known malicious URLs, the status of the message is determined as malicious. The message is released to be displayed along with the status.

Inventors: Gorter; Charlene Moss (Concord, NC), Mayers; Kyle Andrew (Charlotte, NC), Lohar; Sanjay Arjun (Charlotte, NC), Siekman; James (Charlotte, NC), McCormick; Connor Thomas (Westminster, CO)

Applicant: Bank of America Corporation (Charlotte, NC)

Family ID: 92119212

Assignee: Bank of America Corporation (Charlotte, NC)

Appl. No.: 18/164056

Filed: February 03, 2023

Prior Publication Data

Document Identifier

Publication Date

US 20240267410 A1

Aug. 08, 2024

Publication Classification

Int. Cl.: H04L9/40 (20220101)

U.S. Cl.:

CPC H04L63/1483 (20130101); H04L63/1425 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8255572	12/2011	Coomer	N/A	N/A
8510829	12/2012	Mohandas et al.	N/A	N/A
8549642	12/2012	Lee	N/A	N/A
8805996	12/2013	Gauvin	N/A	N/A
9104867	12/2014	Thioux et al.	N/A	N/A
9223980	12/2014	Bao	N/A	N/A
9241009	12/2015	Starink et al.	N/A	N/A
9356948	12/2015	Higbee et al.	N/A	N/A
9565202	12/2016	Kindlund et al.	N/A	N/A
9876753	12/2017	Hawthorn	N/A	N/A
9888016	12/2017	Amin et al.	N/A	N/A
10025927	12/2017	Khalid et al.	N/A	N/A
10027690	12/2017	Aziz	N/A	G06F 21/56
10192052	12/2018	Singh et al.	N/A	N/A
10505956	12/2018	Pidathala et al.	N/A	N/A
10523609	12/2018	Subramanian	N/A	N/A
10616272	12/2019	Chambers et al.	N/A	N/A
11483343	12/2021	Kohavi	N/A	N/A
12143418	12/2023	Gopathy	N/A	H04L 63/0281
2016/0366100	12/2015	Liu et al.	N/A	N/A
2021/0097168	12/2020	Patel	N/A	G06F 21/554
2022/0070216	12/2021	Kohavi	N/A	N/A
2024/0338447	12/2023	Burns	N/A	G06F 21/53

Primary Examiner: Swearingen; Jeffrey R

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to detecting malicious content, and more specifically to a system and method for analyzing messages for malicious content using a cloud computing

system.

BACKGROUND

(2) Malicious actors devise various methods to compromise security of computer systems. For example, malicious actors may send to a user a message with a uniform resource locator (URL), which points to a malicious content, such as a malicious file or a malicious webpage. By accessing malicious files or malicious webpages, malicious users may gain unauthorized access to a device of the user and to computer systems that the user has access to.

SUMMARY

(3) The system described in the present disclosure provides several practical applications and technical advantages that overcome the current technical problems with analyzing messages for malicious content.

(4) In general, a client application of a user device may intercept a message and send it to a cloud computing system for analysis. The cloud computing system may be configured to analyze the message received by the user device and determine a status of the message as safe, suspicious, or malicious. The message may comprise a uniform resource locator (URL) that points to a content (e.g., a file or a webpage), which may be safe, suspicious, or malicious. The cloud computing system may determine the status of the message by comparing the URL with known safe and malicious URLs, comparing a source of the message to known safe and malicious sources, and/or analyzing the URL and the content identified by the URL for malicious content and/or spelling errors. After determining the status of the message, the cloud computing system sends the status to the client application of the user device. The client application then releases the message to be displayed by the user device along with this status. The cloud computing system may receive feedback from the user and may update the known safe and malicious URLs and the known safe and malicious sources based on the feedback.

(5) By analyzing a message for the malicious content and providing a user with a status of the message, the user will be alerted to potential dangers of the message and may decide to respond to the message based on the status. For example, if the status of the message is malicious, the user may decide not to access content identified by the URL and malicious users will be denied access to the user device. Accordingly, the security of the user device, and the computing system that the user has access to, is improved. Furthermore, by updating the known safe and malicious URLs and the known safe and malicious sources based on user's feedback most up-to-date information is stored in the cloud computing system, which further improves the security of the user device.

(6) Accordingly, the following disclosure is particularly integrated into practical applications of: (1) identifying malicious content in messages received by a user device; (2) updating the cloud computing system, such that the cloud computing system stores most up-to-date information that is used for analyzing the messages; (3) improving security of the user device; and (4) improving security of a computing system that the user device has authorized access to.

(7) In one embodiment, a system includes a user device and the cloud computing system communicatively coupled to the user device. The user device includes a first memory and a first processor communicatively coupled to the first memory. The first memory is configured to store a client application. The first processor, when executing the client application, is configured to intercept a message comprising a quick response (QR) code or a uniform resource locator (URL), wherein the QR code corresponds to the URL, send the message to a cloud computing system, receive a status of the message from the cloud computing system, and release the message to be displayed along with the status. The cloud computing system includes a second memory and a second processor communicatively coupled to the second memory. The second memory is configured to store known safe URLs, known malicious URLs, known safe sources, known malicious sources, known malicious codes, and hashes of known malicious files. The second processor is configured to receive the message from the user device and determine the status of the message. Determining the status of the message includes determining a source of the message. The

source of the message is compared to the known malicious sources. In response to the source of the message not matching any of the known malicious sources, the source of the message is compared to the known safe sources. In response to the source of the message not matching any of the known safe sources, the URL is compared to the known malicious URLs. In response to the URL matching a respective one of the known malicious URLs, the status of the message is determined as malicious. The status of the message is sent to the user device.

(8) Certain embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, where like reference numerals represent like parts.

(2) FIG. 1 illustrates an embodiment of a system configured to analyze messages for malicious content; and

(3) FIGS. 2A, 2B, and 2C illustrate an example operational flow of system of FIG. 1 for analyzing messages for malicious content.

DETAILED DESCRIPTION

(4) As described above, previous technologies fail to provide effective solutions for analyzing messages for malicious content. Embodiments of the present disclosure and their advantages may be understood by referring to FIGS. 1, 2A, 2B, and 2C. FIGS. 1, 2A, 2B, and 2C are used to describe a system and method for analyzing messages for malicious content.

System Overview

(5) FIG. 1 illustrates an embodiment of a system **100** that is generally configured analyze messages for malicious content. In certain embodiments, the system **100** comprises a user device **106** operably coupled to a cloud computing system **140** via a network **102**. The system **100** may be further operably coupled to an external system **178** via the network **102**. Network **102** enables the communication between the components of the system **100**. In other embodiments, system **100** may not have all the components listed and/or may have other elements instead of, or in addition to, those listed above.

(6) In general, a client application **120** of a user device **106** may intercept a message and send it to a cloud computing system **140** for analysis. The cloud computing system **140** may be configured to analyze the message received by the user device **106** and determine a status of the message as safe, suspicious, or malicious. The message may comprise a uniform resource locator (URL) that points to a content (e.g., a file or a webpage), which may be safe, suspicious, or malicious. The cloud computing system **140** may determine the status of the message by comparing the URL with known safe and malicious URLs, comparing a source of the message to known safe and malicious sources, and/or analyzing the URL and the content identified by the URL for malicious content and/or spelling errors. After determining the status of the message, the cloud computing system **140** sends the status to the client application **120** of the user device **106**. The client application **120** then releases the message to be displayed by the user device **106** along with this status. The cloud computing system **140** may receive feedback from the user and may update the known safe and malicious URLs and the known safe and malicious sources based on the feedback.

System Components

Network

(7) Network **102** may be any suitable type of wireless and/or wired network. The network **102** may

or may not be connected to the Internet or public network. The network **102** may include all or a portion of an Intranet, a peer-to-peer network, a switched telephone network, a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), a wireless PAN (WPAN), an overlay network, a software-defined network (SDN), a virtual private network (VPN), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a plain old telephone (POT) network, a wireless data network (e.g., WiFi, WiGig, WiMax, etc.), a long-term evolution (LTE) network, a universal mobile telecommunications system (UMTS) network, a peer-to-peer (P2P) network, a Bluetooth network, a near field communication (NFC) network, and/or any other suitable network. The network **102** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

User Device

(8) User device **106** is generally any device that is configured to process data and interact with a user **104**. Examples of the user device **106** include, but are not limited to, a personal computer, a desktop computer, a workstation, a server, a laptop, a tablet computer, a mobile phone (such as a smartphone), etc. The user device **106** may include a user interface, such as a display, a microphone, keypad, or other appropriate terminal equipment usable by the user **104**.

(9) The device **106** may comprise a processor **108** in signal communication with a memory **116** and a network interface **110**. Processor **108** comprises one or more processors operably coupled to the memory **110**. The processor **108** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **108** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **108** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The one or more processors are configured to implement various software instructions to perform the operations described herein. For example, the one or more processors are configured to execute software instructions **118** and perform one or more functions described herein.

(10) Network interface **110** is configured to enable wired and/or wireless communications (e.g., via network **102**). The network interface **110** is configured to communicate data between the user device **106** and other components of the system **100**. For example, the network interface **110** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **108** is configured to send and receive data using the network interface **110**. The network interface **110** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(11) Memory **116** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **116** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. The memory **116** may store any of the information described in FIGS. **1**, **2A**, **2B**, and **2C** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by the processor **108**. The memory **116** is operable to store software instructions **118**, and/or any other data and instructions. The software instructions **118** may comprise any suitable set of software instructions, logic, rules, or code operable to be executed by the processor **108**.

(12) The user device **106** may be configured to receive a message (e.g., message **128** or **132**) or a quick response (QR) code (e.g., QR code **122**). The message **126** may comprise a QR code **128**, which may encode a URL **130**. The message **132** may comprise a URL **134**. The messages **126** and **132** may be text messages, short messaging service (SMS) messages, or the like. The user device **106** may receive the messages **126** and **132** from legitimate and/or malicious sources. The user

device **106** may further comprise a display **112**, which may be configured to display received messages. The QR code **122** may encode a URL **124**. The user device **106** may comprise a camera **114**, which may be configured to capture the QR code **122**. The QR code **122** may be from legitimate and/or malicious sources.

(13) The memory **116** may further store a client application **120** that are operable to implement the function(s) described herein when executed by the processor **108**. In certain, embodiments, the processor **108**, when executing the client application **120**, is configured to intercept a message (e.g., message **128** or **132**) or a QR code (e.g., QR code **122**) that is received by the user device **106**. The processor **108** is configured to send the message or the QR code to the cloud computing system **140**. The processor **108** receives a status (e.g., status **138**) of the message or the QR code from the cloud computing system **140**, and releases the message or the QR code to be displayed along with the status. The user **104** may respond to the received message based on the status. The user **104** may ignore or delete the message if the status is identified as malicious. The user **104** may ignore or delete the message or follow the URL of the message if the status is identified as suspicious. The user **104** may follow the URL of the message if the status is identified as safe.

(14) The processor **108** may be further configured to receive feedback **136** from the user **104** and send the feedback **136** to the cloud computing system **140**. In one embodiment, the feedback **136** may comprise an information that a source that was previously identified as safe is actually malicious. In other embodiment, the feedback **136** may comprise an information that the source that was previously identified as suspicious is actually malicious. In yet other embodiment, the feedback **136** may comprise an information that the source that was previously identified as suspicious is actually safe.

(15) In one embodiment, the feedback **136** may comprise an information that a URL that was previously identified as safe is actually malicious. In other embodiment, the feedback **136** may comprise an information that the URL that was previously identified as suspicious is actually malicious. In yet other embodiment, the feedback **136** may comprise an information that the URL that was previously identified as suspicious is actually safe.

External System

(16) External system **178** is generally any device that is configured to process data and communicate with components of the system **100** via the network **102**. The external system **178** may comprise a processor **180** in signal communication with a memory **184** and a network interface **182**.

(17) Processor **180** comprises one or more processors operably coupled to the memory **184**. The processor **180** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **180** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **180** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The one or more processors are configured to implement various software instructions to perform the operations described herein. For example, the one or more processors are configured to execute software instructions **186** and perform one or more functions described herein.

(18) Network interface **182** is configured to enable wired and/or wireless communications (e.g., via network **102**). The network interface **182** is configured to communicate data between the external system **178** and components of the system **100**. For example, the network interface **182** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **180** is configured to send and receive data using the network interface **182**. The network interface **182** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(19) Memory **184** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **184** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. The memory **184** is operable to store software instructions **186**, and/or any other data and instructions. The software instructions **186** may comprise any suitable set of software instructions, logic, rules, or code operable to be executed by the processor **180**.

(20) The external system **178** may be configured to host files (e.g., file **172**) and/or webpages (e.g., webpage **174**). In such embodiments, the memory **184** may further store the files and/or the webpages. As described below in greater detail, the files and/or the webpages may be retrieved from the external system **178** by the cloud computing system **140** during operation.

Cloud Computing System

(21) The cloud computing system **140** is generally any device that is configured to process data and communicate with other components of the system **100** via the network **102**. The cloud computing system **140** may comprise a processor **142** in signal communication with a memory **146** and a network interface **144**.

(22) Processor **142** comprises one or more processors operably coupled to the memory **146**. The processor **142** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **142** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **142** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The one or more processors are configured to implement various software instructions to perform the operations described herein. For example, the one or more processors are configured to execute software instructions **148** and perform one or more functions described herein.

(23) Network interface **144** is configured to enable wired and/or wireless communications (e.g., via network **102**). The network interface **144** is configured to communicate data between the cloud computing system **140** and other components of the system **100**. For example, the network interface **144** may comprise a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a modem, a switch, or a router. The processor **142** is configured to send and receive data using the network interface **144**. The network interface **144** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(24) Memory **146** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). Memory **146** may be implemented using one or more disks, tape drives, solid-state drives, and/or the like. The memory **146** may store any of the information described in FIGS. **1**, **2A**, **2B**, and **2C** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein. The memory **146** is operable to store software instructions **148**, and/or any other data and **10** instructions. The software instructions **148** may comprise any suitable set of software instructions, logic, rules, or code operable to be executed by the processor **142**.

(25) Memory **146** may be operable to store a database **150**. The database **150** may store various parameters that are used by the cloud computing system **140** to analyze received messages and QR codes. In certain embodiments, the memory **146** stores known safe sources **152**, known malicious sources **154**, known safe URLs **156**, known malicious URLs **158**, hashes **160** of known malicious files, and known malicious codes **162**.

(26) At operation, the cloud computing system **140** is configured to receive a message (e.g.,

message **128** or **132**) or a QR code (e.g., QR code **122**) from the user device **106**. Subsequently, the cloud computing system **140** determines a source **164** of the message or the QR code. In case of the message, the cloud computing system **140** may determine the source **164** of the message based on a phone number from which the message was sent. In case of the QR code, the cloud computing system **140** may determine the source **164** of the QR code based on surroundings that are captured by the camera **114** while capturing the QR code. In other embodiments, geotagging information of an image comprising the QR code may be used to determine the source **164** of the QR code.

(27) The cloud computing system **140** compares the source **164** to the known malicious sources **154**. In response to determining the source **164** matches one of the known malicious sources **154**, the cloud computing system **140** determines the status **138** of the message or the QR code as malicious. In response to determining that the source **164** does not match any of the known malicious sources **154**, the cloud computing system **140** compares the source **164** to the known safe sources **152**. In response to determining that the source **164** matches one of the known safe sources **152**, the cloud computing system **140** determines the status **138** of the message or the QR code as safe. In response to determining that the source **164** does not match any of the known safe sources **152**, the cloud computing system **140** compares a URL (e.g., URLs **124**, **130** and **134**) of the messages or the QR code to the known malicious URLs **158**. In response to determining that the URL matches one of the known malicious URLs **158**, the cloud computing system **140** determines the status **138** of the message or the QR code as malicious. In response to determining that the URL does not match any of the known malicious URLs **158**, the cloud computing system **140** compares the URL to the known safe URLs **156**. In response to determining that the URL matches one of the known safe URLs **156**, the cloud computing system **140** determines the status **138** of the message or the QR code as safe.

(28) In response to determining the URL does not match any of the known safe URLs **156**, the cloud computing system **140** analyzes the URL for spelling errors **168**. In response to identifying at least one spelling error, the cloud computing system **140** determines the status **138** of the message or the QR code as suspicious. In response to identifying no spelling error, the cloud computing system **140** determines a destination **170** that corresponds to the URL. In certain embodiments, the destination **170** may be a file **172** or a webpage **174** hosted by an external system **178**. The file may be a text file, an image file, or the like.

(29) In case when the cloud computing system **140** determines that the destination **170** is the file **172** hosted by the external system **178**, the cloud computing system **140** retrieves the file **172** from the external system **178** and generates a hash **166** of the file **172**. Subsequently, the cloud computing system **140** compares the hash **166** to the hashes **160** of known malicious files. In response to determining that the hash **166** matches one of the hashes **160**, the cloud computing system **140** determines the status **138** of the message or the QR code as malicious.

(30) In response to determining that the hash **166** does not match any of the hashes **160**, the cloud computing system **140** analyzes the file **172** for spelling errors **168**. In response to identifying at least one spelling error, the cloud computing system **140** determines the status **138** of the message or the QR code as suspicious. In response to identifying no spelling error, the cloud computing system **140** determines the status **138** of the message or the QR code as safe.

(31) In case when the cloud computing system **140** determines that the destination **170** is the webpage **174** hosted by the external system **178**, the cloud computing system **140** retrieves the webpage **174** from the external system **178** and analyzes a source code **176** of the webpage **174** for malicious codes **162**. In response to identifying at least one malicious code, the cloud computing system **140** determines the status **138** of the message or the QR code as malicious.

(32) In response to identifying no malicious code, the cloud computing system **140** analyzes contents of the webpage **174** for spelling errors. In response to identifying at least one spelling error, the cloud computing system **140** determines the status **138** of the message or the QR code as suspicious. In response to identifying no spelling error, the cloud computing system **140** determines

the status **138** of the message or the QR code as safe.

(33) In certain embodiments, the cloud computing system **140** may be further configured to receive feedback **136** from the user device **160** and update the known safe sources **152**, the known malicious sources **154**, the known safe URLs **156**, the known malicious URLs **158**, the hashes **160** of known malicious files, and the known malicious codes **162** based on the feedback **136**.

(34) In one embodiment, the feedback **136** may comprise an information that a source that was previously identified as safe is actually malicious. Such a source is then removed from the known safe source **152**. In other embodiment, the feedback **136** may comprise an information that the source that was previously identified as suspicious is actually malicious. Such a source is then added to the known malicious sources **154**. In yet other embodiment, the feedback **136** may comprise an information that the source that was previously identified as suspicious is actually safe. Such a source is then added to the known safe sources **152**.

(35) In one embodiment, the feedback **136** may comprise an information that a URL that was previously identified as safe is actually malicious. Such a URL is then removed from the known safe URLs **156**. In other embodiment, the feedback **136** may comprise an information that the URL that was previously identified as suspicious is actually malicious. Such a URL is then added to the known malicious URLs **158**. In yet other embodiment, the feedback **136** may comprise an information that the URL that was previously identified as suspicious is actually safe. Such a URL is then added to the known safe URLs **156**.

Example Method for Identifying Security Threats of an Application

(36) FIGS. 2A, 2B, and 2C illustrate an example flowchart of a method **200** for analyzing messages for malicious content using a cloud computing system. Modifications, additions, or omissions may be made to method **200**. Method **200** may include more, fewer, or other operations. For example, operations may be performed in parallel or in any suitable order. For example, one or more operations of method **200** may be implemented, at least in part, in the form of the software instructions (e.g., instructions **118**, and/or **148** of FIG. 1), stored on non-transitory, tangible, machine-readable medium (e.g., memories **116**, and/or **146** of FIG. 1) that when executed by one or more processors (e.g., processors **108**, and/or **142** of FIG. 1) may cause the one or more processors to perform operations **202-270**.

(37) At operation **202**, a client application (e.g., client application **120** of FIG. 1) running on a user device (e.g., user device **106** of FIG. 1) intercepts a message (e.g., messages **126** and **132** of

(38) FIG. 1) comprising a QR code (e.g., QR code **128** of FIG. 1) or a URL (e.g., URLs **130** and **134** of FIG. 1), such that the QR code corresponds to the URL.

(39) At operation **204**, the user device sends the message **132** to a cloud computing system (e.g., cloud computing system **140** of FIG. 1).

(40) At operation **206**, the cloud computing system **140** receives the message **132** from the user device **106**.

(41) At operation **208**, the cloud computing system **140** determines a source (e.g., source **164** of FIG. 1) of the message **132**.

(42) At operation **210**, the cloud computing system **140** compares the source **164** to known malicious sources (e.g., known malicious sources **154** of FIG. 1).

(43) At operation **212**, the cloud computing system **140** determines if the source **164** matches a known malicious source **154**.

(44) In response to the source **164** matching the known malicious source **154**, method **200** continues to operation **248**, where the cloud computing system **140** determines a status (e.g., status **138** of FIG. 1) of the message **132** as malicious.

(45) In response to the source **164** not matching the known malicious **154** source as determined at operation **212**, method **200** continues to operation **214**, where the cloud computing system **140** compares the source **164** to known safe sources (e.g., known safe sources **152** of FIG. 1).

(46) At operation **216**, the cloud computing system **140** determines if the source **164** matches a

known safe source **152**.

(47) In response to the source **164** matching the known safe source **152**, method **200** continues to operation **246**, where the cloud computing system **140** determines the status **138** of the message **132** as safe.

(48) In response to the source **164** not matching the known safe source **152** as determined at operation **216**, method **200** continues to operation **218**, where the cloud computing system **140** compares the URL **134** to known malicious URLs (e.g., known malicious URLs **158** of FIG. 1).

(49) At operation **220**, the cloud computing system **140** determines if the URL **134** matches a known malicious URL **158**.

(50) In response to the URL **134** matching the known malicious URL **158**, method **200** continues to operation **248**, where the cloud computing system **140** determines the status **138** of the message **132** as malicious.

(51) In response to the URL **134** not matching the known malicious URL **158** as determined at operation **220**, method **200** continues to operation **222**, where the cloud computing system **140** compares the URL **134** to known safe URLs (e.g., known safe URLs **156** of FIG. 1).

(52) At operation **224**, the cloud computing system **140** determines if URL **134** matches a known safe URL **156**.

(53) In response to the URL **134** matching the known safe URL **156**, method **200** continues to operation **246**, where the cloud computing system **140** determines the status **138** of the message **132** as safe.

(54) In response to the URL **134** not matching the known safe URL **156** as determined at operation **224**, method **200** continues to operation **226**, where the cloud computing system **140** analyzes the URL **134** for spelling errors (e.g., spelling errors **168** of FIG. 1).

(55) At operation **228**, the cloud computing system **140** determines if at least one spelling error **168** is identified.

(56) In response to identifying at least one spelling error **168** as determined at operation **228**, method **200** continues to operation **250**, where the cloud computing system **140** determines the status **138** of the message **132** as suspicious.

(57) In response to identifying no spelling error as determined at operation **228**, method **200** continues to operation **230**, where the cloud computing system **140** determines a destination (e.g., destination **170** of FIG. 1) corresponding to the URL **134**.

(58) At operation **232**, the cloud computing system **140** determines if the destination **170** is a file (e.g., file **172** of FIG. 1) or a webpage (e.g., webpage **174** of FIG. 1).

(59) In response to determining that the destination **170** is the file **172** hosted by an external system (e.g., external system **178** of FIG. 1), method **200** continues to operation **234**, where the cloud computing system **140** retrieves the file **172** from the external system **178**.

(60) At operation **236**, the cloud computing system **140** generates a hash (e.g., hash **166** of FIG. 1) of the file **172**.

(61) At operation **238**, the cloud computing system **140** compares the hash to hashes (e.g., hash **160** of FIG. 1) of malicious files.

(62) At operation **240**, the cloud computing system **140** determines if the hash **166** matches a hash **160** of a known malicious file.

(63) In response to determining that the hash **166** matches the hash **160** of the known malicious file as determined at operation **240**, method **200** continues to operation **248**, where the cloud computing system **140** determines the status **138** of the message **132** as malicious.

(64) In response to determining that the hash **166** does not match the hash **160** of the known malicious file as determined at operation **240**, method **200** continues to operation **242**, where the cloud computing system **140** analyzes the file **172** for spelling errors **168**.

(65) At operation **244**, the cloud computing system **140** determines if at least one spelling error **168** is identified.

(66) In response to identifying at least one spelling error **168** as determined at operation **244**, method **200** continues to operation **250**, where the cloud computing system **140** determines the status **138** of the message **132** as suspicious.

(67) In response to identifying no spelling error as determined at operation **244**, method **200** continues to operation **246**, where the cloud computing system **140** determines the status **138** of the message **132** as safe.

(68) In response to determining at operation **232** that the destination **170** is the webpage **174** hosted by the external system **178**, method **200** continues to operation **262**, where the cloud computing system **140** retrieves the webpage **174** from the external system **178**.

(69) At operation **264**, the cloud computing system **140** analyzes a source code (e.g., source code **176** of FIG. 1) of the webpage **174** for malicious codes (e.g., malicious codes **162** of FIG. 1).

(70) At operation **266**, the cloud computing system **140** determines if at least one malicious code **162** is identified.

(71) In response to identifying at least one malicious code **162** as determined at operation **266**, the method **200** continues to operation **248**, where the cloud computing system **140** determines the status **138** of the message **132** as malicious.

(72) In response to identifying no malicious code as determined at operation **266**, the method **200** continues to operation **268**, where the cloud computing system **140** analyzes contents of the webpage **174** for spelling errors **168**.

(73) At operation **270**, the cloud computing system **140** determines if at least one spelling error **168** is identified.

(74) In response to identifying at least one spelling error **168** as determined at operation **270**, method **200** continues to operation **250**, where the cloud computing system **140** determines the status **138** of the message **132** as suspicious.

(75) In response to identifying no spelling error as determined at operation **270**, method **200** continues to operation **246**, where the cloud computing system **140** determines the status **138** of the message **132** as safe.

(76) After determining the status **138** of the message **132**, method **200** continues to operation **252**, wherein the client application **120** running on the user device **106** receives the status **138** of the message **132** from the cloud computing system **140**.

(77) At operation **254**, the client application **120** running on the user device **106** releases the message **132** to be displayed along with the status **138**. In certain embodiments, the message **132** along with the status **138** is displayed by a display (e.g., display **112** of FIG. 1) of the user device **106**.

(78) At operation **256**, the client application **120** running on the user device **106** receives feedback (e.g., feedback **136** of FIG. 1) from the user **104**.

(79) At operation **258**, the cloud computing system **140** receives the feedback **136** from the user device **106**.

(80) At operation **260**, the cloud computing system **140** updates the known safe sources **152**, the known malicious sources **154**, the known safe URLs **156**, the known malicious URLs **158**, the hashes **160** of known malicious files, and/or the known malicious codes **160** based on the feedback **136**.

(81) In one embodiment, the feedback **136** may comprise an information that a source that was previously identified as safe is actually malicious. Such a source is then removed from the known safe sources **152**. In other embodiment, the feedback **136** may comprise an information that a source that was previously identified as suspicious is actually malicious. Such a source is then added to the known malicious sources **154**. In yet other embodiment, the feedback **136** may comprise an information that the source that was previously identified as suspicious is actually safe. Such a source is then added to the known safe sources **152**.

(82) In one embodiment, the feedback **136** may comprise an information that a URL that was

previously identified as safe is actually malicious. Such a URL is then removed from the known safe URLs **156**. In other embodiment, the feedback **136** may comprise an information that the URL that was previously identified as suspicious is actually malicious. Such a URL is then added to the known malicious URLs **158**. In yet other embodiment, the feedback **136** may comprise an information that the URL that was previously identified as suspicious is actually safe. Such a URL is then added to the known safe URLs **156**.

(83) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented.

(84) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(85) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a user device comprising: a first memory configured to store a client application; and a first processor communicatively coupled to the first memory, wherein the first processor, when executing the client application, is configured to: intercept a message comprising a quick response (QR) code or a uniform resource locator (URL), wherein the QR code corresponds to the URL; send the message to a cloud computing system; receive a status of the message from the cloud computing system; and release the message to be displayed along with the status; and the cloud computing system communicatively coupled to the user device, wherein the cloud computing system comprises: a second memory configured to store: known safe URLs; known malicious URLs; known safe sources; known malicious sources; known malicious codes; and hashes of known malicious files; and a second processor communicatively coupled to the second memory, the second processor configured to: receive the message from the user device; determine the status of the message, wherein determining the status of the message comprises: determining a source of the message; comparing the source of the message to the known malicious sources; in response to the source of the message not matching any of the known malicious sources, comparing the source of the message to the known safe sources; in response to the source of the message not matching any of the known safe sources, comparing the URL to the known malicious URLs; and in response to the URL matching a respective one of the known malicious URLs, determining the status of the message as malicious; in response to the URL not matching any of the known malicious URLs, compare the URL to the known safe URLs; in response to the URL matching a respective one of the known safe URLs, determine the status of the message as safe; in response to the URL not matching any of the known safe URLs, analyze the URL for spelling errors; in response to identifying at least one spelling error in the URL, determine the status of the message as suspicious; in response to identifying no spelling errors in the URL, determine a destination

identified by the URL; in response to determining that the destination identified by the URL is a file, determine a hash of the file; compare the hash of the file to the hashes of known malicious files; and in response to the hash of the file matching a hash of one of the known malicious files, determine the status of the message as malicious; and send the status of the message to the user device.

2. The system of claim 1, wherein determining the status of the message further comprises: in response to the hash of the file not matching any of the hashes of known malicious files, analyzing contents of the file for spelling errors; in response to identifying at least one spelling error in the file, determining the status of the message as suspicious; and in response to identifying no spelling errors in the file, determining the status of the message as safe.

3. The system of claim 2, wherein determining the status of the message further comprises: in response to determining that the destination identified by the URL is a webpage, analyzing a source code of the webpage for malicious codes; and in response to identifying at least one malicious code, determining the status of the message as malicious.

4. The system of claim 3, wherein determining the status of the message further comprises: in response to identifying no malicious code, analyzing contents of the webpage for spelling errors; in response to identifying at least one spelling error in the webpage, determining the status of the message as suspicious; and in response to identifying no spelling errors in the webpage, determine the status of the message as safe.

5. A method comprising: intercepting a message comprising a quick response (QR) code or a uniform resource locator (URL), wherein the QR code corresponds to the URL; determining a status of the message, wherein determining the status of the message comprises: determining a source of the message; comparing the source of the message to known malicious sources; in response to the source of the message not matching any of the known malicious sources, comparing the source of the message to known safe sources; in response to the source of the message not matching any of the known safe sources, comparing the URL to known malicious URLs; and in response to the URL matching a respective one of the known malicious URLs, determining the status of the message as malicious; in response to the URL not matching any of the known malicious URLs, comparing the URL to known safe URLs; in response to the URL matching a respective one of the known safe URLs, determining the status of the message as safe; in response to the URL not matching any of the known safe URLs, analyzing the URL for spelling errors; in response to identifying at least one spelling error in the URL, determining the status of the message as suspicious; in response to identifying no spelling errors in the URL, determining a destination identified by the URL; in response to determining that the destination identified by the URL is a file, determining a hash of the file; comparing the hash of the file to hashes of known malicious files; and in response to the hash of the file matching a hash of one of the known malicious files, determining the status of the message as malicious; and releasing the message to be displayed along with the status.

6. The method of claim 5, further comprising: in response to the hash of the file not matching any of the hashes of known malicious files, analyzing contents of the file for spelling errors; in response to identifying at least one spelling error in the file, determining the status of the message as suspicious; and in response to identifying no spelling errors in the file, determining the status of the message as safe.

7. The method of claim 6, further comprising: in response to determining that the destination identified by the URL is a webpage, analyzing a source code of the webpage for malicious codes; and in response to identifying at least one malicious code, determining the status of the message as malicious.

8. The method of claim 7, further comprising: in response to identifying no malicious code, analyzing contents of the webpage for spelling errors; in response to identifying at least one spelling error in the webpage, determining the status of the message as suspicious; and in response

to identifying no spelling errors in the webpage, determine the status of the message as safe.

9. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to: intercept a message comprising a quick response (QR) code or a uniform resource locator (URL), wherein the QR code corresponds to the URL; determine a status of the message, wherein determining the status of the message comprises: determining a source of the message; comparing the source of the message to known malicious sources; in response to the source of the message not matching any of the known malicious sources, comparing the source of the message to known safe sources; in response to the source of the message not matching any of the known safe sources, comparing the URL to known malicious URLs; and in response to the URL matching a respective one of the known malicious URLs, determining the status of the message as malicious; in response to the URL not matching any of the known malicious URLs, compare the URL to known safe URLs; in response to the URL matching a respective one of the known safe URLs, determine the status of the message as safe; in response to the URL not matching any of the known safe URLs, analyze the URL for spelling errors; and in response to identifying at least one spelling error in the URL, determine the status of the message as suspicious; in response to identifying no spelling errors in the URL, determine a destination identified by the URL; in response to determining that the destination identified by the URL is a file, determine a hash of the file; compare the hash of the file to hashes of known malicious files; and in response to the hash of the file matching a hash of one of the known malicious files, determine the status of the message as malicious; and release the message to be displayed along with the status.

10. The non-transitory computer-readable medium of claim 9, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to: in response to the hash of the file not matching any of the hashes of known malicious files, analyze contents of the file for spelling errors; in response to identifying at least one spelling error in the file, determine the status of the message as suspicious; and in response to identifying no spelling errors in the file, determine the status of the message as safe.

11. The non-transitory computer-readable medium of claim 10, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to: in response to determining that the destination identified by the URL is a webpage, analyze a source code of the webpage for malicious codes; and in response to identifying at least one malicious code, determine the status of the message as malicious.
